Algebra 2: Unit 6 Day 2 — ANSWER KEY

Part 1: Multi-Step Isolating

1. $4\sqrt[3]{x+2}+5=-3$
$4\sqrt[3]{x+2}=-8 \to \sqrt[3]{x+2}=-2$
$x+2=-8$
Answer: $x=-10$

2. $5(x-2)^{\frac{2}{3}}+7=52$
$5(x-2)^{\frac{2}{3}}=45 \to (x-2)^{\frac{2}{3}}=9$
$x-2=\pm 27$
Answer: $x=29,-25$

3. $\frac{2}{3}(x-5)^{\frac{1}{3}}-4=-2$
$\frac{2}{3}(x-5)^{\frac{1}{3}}=2 \to (x-5)^{\frac{1}{3}}=3$
$x-5=27$
Answer: $x=32$

4. $3+2(x-1)^{\frac{1}{3}}=7$
$2(x-1)^{\frac{1}{3}}=4 \to (x-1)^{\frac{1}{3}}=2$
$x-1=8$
Answer: $x=9$

Part 2: Coefficients

1. $3\sqrt[3]{x+5}=6\sqrt[3]{x+4}$
$\sqrt[3]{x+5}=2\sqrt[3]{x+4}$
$x+5=8(x+4) \to x+5=8x+32$
$-27=7x$
Answer: $x=-27/7\approx -3.8$

2. $2\sqrt[3]{10x+1}=3\sqrt[3]{3x-1}$
$8(10x+1)=27(3x-1)$
$80x+8=81x-27$
Answer: $x=35$

Part 3: Applications

1. Volume: $s=\sqrt[3]{729}$
Answer: $s=9$

2. Wind: $54=0.2v^3 \to 270=v^3$
Answer: $v=\sqrt[3]{270}\approx 6.46$

You Try!

1. $3\sqrt[3]{2x+1}=9 \to 2x+1=27$
Answer: $x=13$

2. $4\sqrt[3]{x+20}=-4 \to x+20=-1$
Answer: $x=-21$

3. $(x-3)^{\frac{1}{3}}=\frac{12}{5} \to x-3=\frac{1728}{125}$
Answer: $x=\frac{2103}{125}=16.824$

4. $(3x+5)^{\frac{2}{3}}=9 \to 3x+5=\pm 27$
Answer: $x=\frac{22}{3},-\frac{32}{3}$

5. Volume: $s=\sqrt[3]{64}$
Answer: $s=4$

6. $2\sqrt[3]{x+3}=5 \to x+3=\frac{125}{8}$
Answer: $x=\frac{101}{8}=12.625$

7. $4\sqrt[3]{x+4}=12 \to x+4=27$
Answer: $x=23$

8. $\frac{1}{2}(4x+4)^{\frac{1}{3}}=-2 \to 4x+4=-64$
Answer: $x=-17$

9. $(3x-2)^{\frac{2}{3}}=4 \to 3x-2=\pm 8$
Answer: $x=\frac{10}{3},-2$

10. Volume: $s=\sqrt[3]{15625}$
Answer: $s=25$